

Learning Transferable Visual Models From Natural Language Supervision

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Date of publication: 26 Feb 2021

Summarized by: Suman Dhakal

Problem Statement:

Traditional computer vision models are trained on large labeled dataset (eg. ImageNet) to get the feature representation of the visual components, and these models could be used as a pre-trained model for downstream tasks. Even though these models are good with feature extraction, they need a large amount of label dataset for training and are still missing generality about other visual components.

This paper addresses these problems by building a more general and transferable vision model without requiring task specific supervision.

Objective:

Learning directly from image text (caption) pairs, instead of image - categorical labels, collected from the internet, helps the model to learn broad visual representation and visual component understanding. These pre-trained models enable zero-shot transfer learning.

Zero-shot learning is a machine learning paradigm where a model is trained to recognize and classify objects or concepts it has never seen during training. In traditional machine learning, models are trained on labeled data from all classes they are expected to classify. However, in zero-shot learning, the goal is to enable a model to generalize its knowledge and make predictions for classes that were not part of the training data.

Methodology

Past work involves weak supervised models (only a small portion of data is labeled) were trained on millions and billions of images for years whereas recent natural language supervision models (VirTex, ConVIRT) are trained on 100k to 200k for a day. CLIP closed the gap by training on a large scale of data.

- **Trained on 400 million image–text pairs** collected from the web, and named it WIT for WebImageText
- Introduced a **CLIP** (Contrastive Language-Image Pre-training) model which is a simplified version of ConVIRT

First approach: (predictive objective)

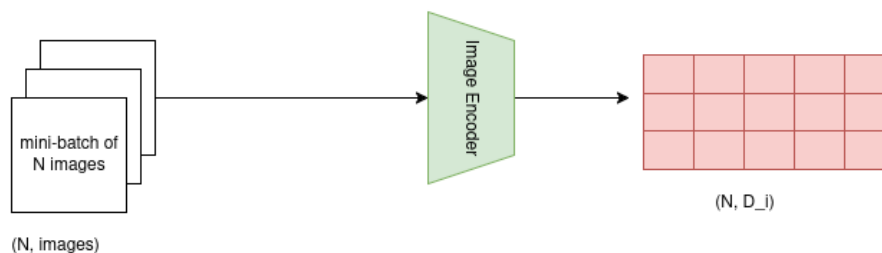
- Similar to VirTex, trained an image CNN and text transformer from scratch to predict the caption of an image
- It is hard to predict the exact word of the text accompanying each image

Second approach: (contrastive objective)

- Given a batch of N (image, text) pairs, CLIP is trained to predict which of $N \times N$ possible image text pairing across the batch actually occurred.
- For image encoder: (*trained from scratch*)
 - ResNet (CNN based) and ViT based model is used
- For text encoder: (*trained from scratch*)
 - Transformer based architecture is used
- Images are passed to the image encoder, and texts are passed in the text encoders, CLIP learns by maximizing the cosine similarity of the embeddings of N real (text, image) pairs in a batch while minimizing the cosine similarity of the embeddings of $N*N - N$ (i.e all pairs - real pairs)
- Symmetric cross entropy loss function is used for model training

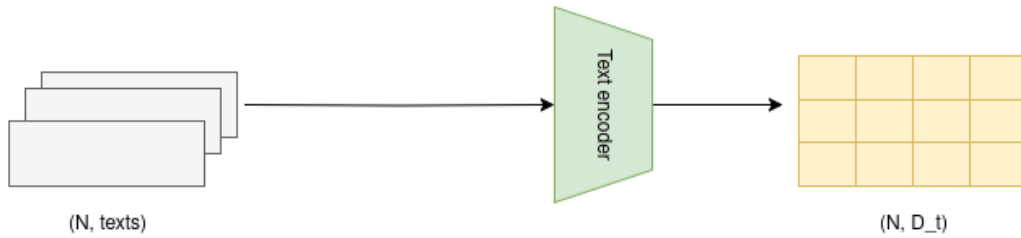
Core Implementation of CLIP:

- **Step 1:**
calculate the feature representations of mini batch of N images



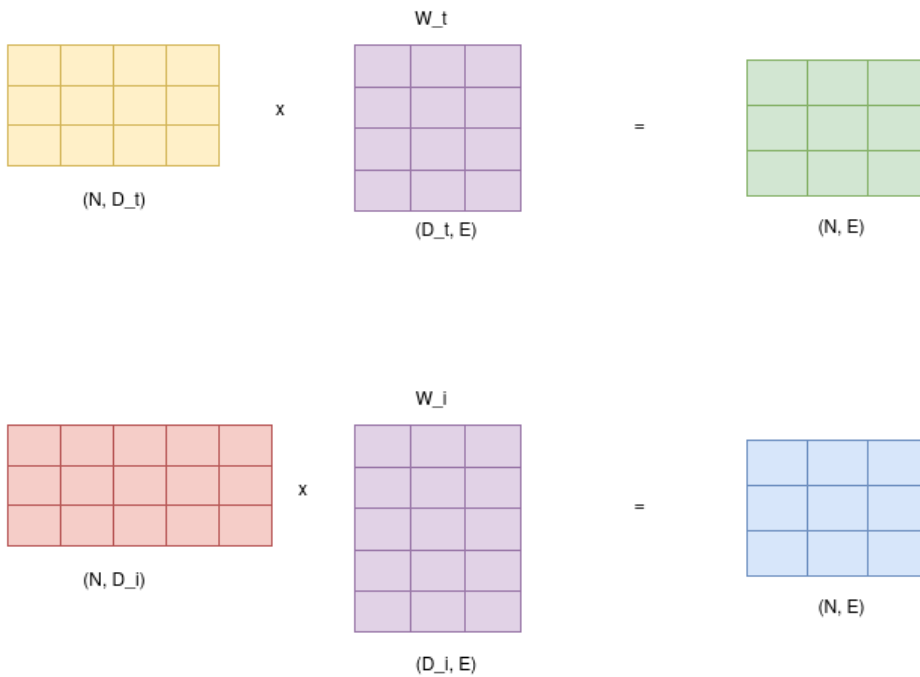
where,
 N is the total number of images in a batch
 D_i is the features vector of an image

- **Step 2:**
Calculate the feature representations of mini batch of N corresponding texts



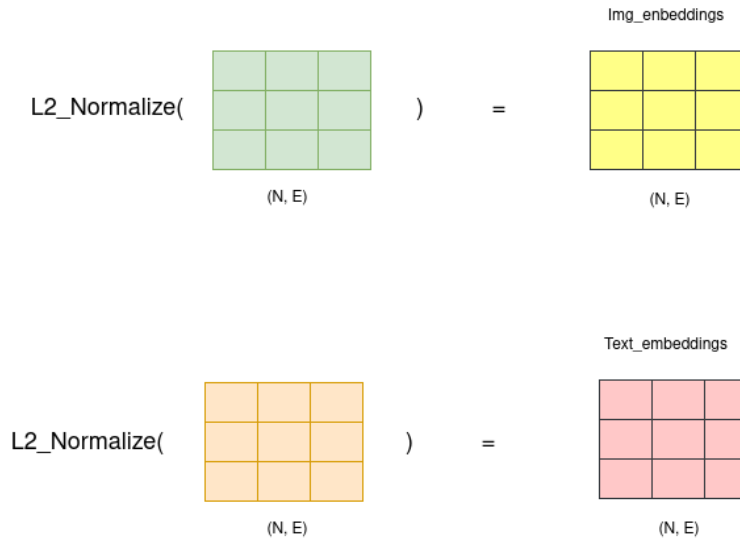
- **Step 3:**

Now perform linear projection to the image and text feature representation to project them into a same vector space



- **Step 4:**

Now we are going to L2 normalize the projected matrix, so that
 $\| \text{image_embedding} \| = \| \text{text_embedding} \| = 1$



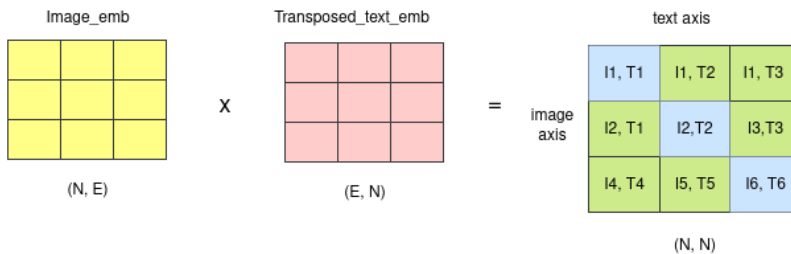
- **Step 5:**

Now let's calculate the cosine similarity between image embeddings and text embeddings

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|}$$

After L2 normalization, we get $\| \text{image_embedding} \| = \| \text{text_embedding} \| = 1$. Thus,

$$\cos \theta = \frac{\text{imageEmbedding} \cdot \text{TextEmbedding}}{1}$$



Note, diagonal element are the correct text image pairs thus, should have highest cosine similarity ideally 1.

- **Step 6:**

Now we are going to calculate cross entropy loss in row wise and column wise:

- Row i : similarity of **image i** with all texts.
- Column j : similarity of **text j** with all images.

1. Image → Text loss

$$L_{i2t} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(\text{image}_i, \text{text}_i))}{\sum_{j=1}^N \exp(\text{sim}(\text{image}_i, \text{text}_j))}$$

Numerator = similarity of (image i , text i).

Denominator = sum over all texts for that image (row of the logits matrix).

2. Text → Image loss

$$L_{t2i} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(\text{text}_i, \text{image}_i))}{\sum_{j=1}^N \exp(\text{sim}(\text{text}_i, \text{image}_j))}$$

Numerator = same similarity value (dot product is symmetric if embeddings are normalized).

Denominator = sum over all images for that text (column of the logits matrix).

Together, CLIP averages them to make training symmetric:

$$\text{loss} = (\text{loss}_{i2t} + \text{loss}_{t2i}) / 2$$

Note that, although the numerator (true pair similarity) is the same in both formulas, the denominator differs.

Pseudo code of above steps:

```
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter
# extract feature representations of each modality

I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

Key Results:

- CLIP shows strong **zero-shot transfer** capabilities: without fine-tuning, it can perform classification by comparing class names with image embeddings.
- Achieves competitive performance on **over 30 benchmark datasets** compared to fully supervised models.

- Outperforms ImageNet-trained models in terms of generalization and robustness across diverse tasks.
- Prompt engineering and ensembling improve zero-shot performance
- Zero-shot CLIP matches the average performance of a 4-shot linear classifier trained on the same feature space and nearly matches the best results of a 16-shot linear classifier across publicly available models
- Fitting a linear classifier on CLIP's features outperforms using the Noisy Student EfficientNet-L2 on 21 out of 27 datasets.
- Few-shot CLIP also increases effective robustness compared to existing ImageNet models but is less robust than zero-shot CLIP
- Zero-shot CLIP is much more robust to distribution shift than standard ImageNet models.
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Conclusion & Contributions:

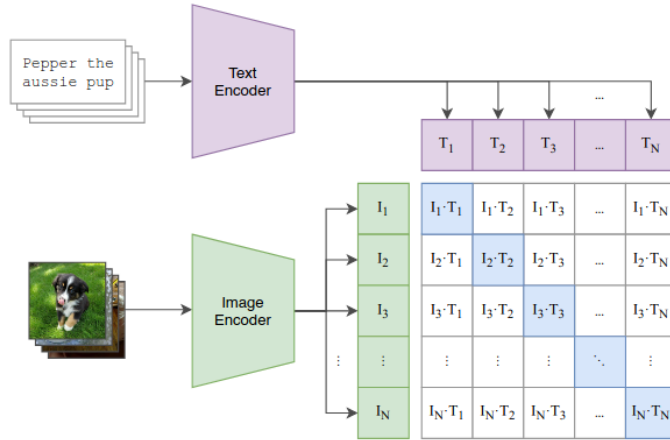
CLIP demonstrates that natural language supervision can replace traditional labeled datasets for vision tasks. It enables scalable pretraining of vision models using freely available web data, making them highly generalizable and adaptable to unseen tasks.

Limitations & Future Work:

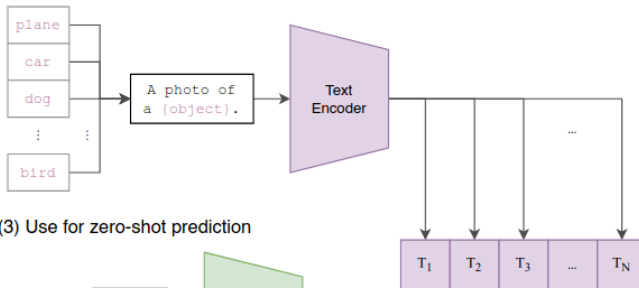
- The model inherits biases from internet data, leading to fairness and ethical concerns.
- Zero-shot performance is strong but still behind task-specific fine-tuned models in certain domains.
- Future research can improve data curation and mitigate bias issues.

Appendices

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

